AI Assignment 2 – Debrief

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1. Kaito: Why the Climb stopped

Kaito takes three steps: (0,0)=52 -> (0,1)=58 -> (1,1)=64 -> (2,1)=70. From (2,1), the neighbours in up, right, down, left order are (1,1)=64, (2,2)=64, (3,1)=64 and (2,0)=64. None is higher, so hill climbing stops. The value 70 is a local maximum. The true peak is (6,6)=95, but reaching it requires going downhill first or starting somewhere else.

With the four prescribed starts, random restart reaches 95 for the first time from (7,0). The four climbs take 21 steps in total.

2. Gennai: What changes when the Starting Heat is 40

At the required starting temperature of 20.0, seed 2 finishes at (6,6) with final and best value 95 and 36 accepted moves. Eight of the twenty seeds finish at 95: 2,3,6,7,14,16,18,20.

A downhill move is accepted with probability exp(delta/T). For a drop from 70 to 64, this is about 0.741 at T=20 and 0.861 at T=40. More heat makes it easier to escape the first hill. Multiplying T by 0.97 each iteration gradually makes the search less willing to go downhill. Cooling too quickly can leave it trapped, while cooling too slowly can leave it wandering when the 200 iterations run out.

I repeated the same experiment with only the starting temperature changed to 40.0. Seed 2 still reaches (6,6) and value 95, but accepts 54 moves. Twelve seeds finish at 95: 2,3,4,7,8,10,13,14,16,18,19,20. That raises the success rate from 40% to 60% for these twenty runs. It does not help every seed. Seed 6 succeeds at 20 but fails at 40. Different decisions also change the path and when extra random draws are needed. The submitted Owl Cipher still uses the required temperature of 20.0.

3. Rin: Why the Seeds stopped at 234

The best fitness at generations 0,10,20,30,40,50 is 160,222,234,234,234,234. The GA first reaches its final best in generation 15. The final population average is 148.20, including infeasible pouches with fitness zero.

The GA pouch is 100111100110001, with usefulness 234, load 83 and volume 43. The exact optimum is 101110100100101, with usefulness 241, load 84 and volume 44. Both strings show bit 0 first. The comparison below lists every tool in index order.

S. No.	Tool	GA Bit	Optimum Bit
0	Kunai set	1	1
1	Wire spool	0	0
2	Smoke bombs	0	1
3	Paper bombs	1	1
4	Energy pills	1	1
5	Flash tags	1	0
6	Messenger scroll	1	1
7	Giant shuriken	0	0
8	Spare rations	0	0
9	Wind shuriken	1	1
10	Twin blades	1	0
11	Caltrops	0	0
12	Medical kit	0	1
13	Grappling hook	0	0
14	Binding tags	1	1

The better pouch replaces Flash tags and Twin blades with Smoke bombs and Medical kit. This gains 7 usefulness for 1 extra load and 1 extra volume, staying within both limits. Checking all 32,768 pouches confirms that 241 is optimal.

A population of twenty over fifty generations only samples the possibilities. It cannot guarantee the optimum. Tournament selection and elitism keep good pouches, but can also reduce diversity. This improvement changes four bits together, and intermediate changes can be less useful or infeasible. One possible improvement is a local search that tries removing up to two tools and adding up to two others while checking both limits. It would find this exchange. I have left the prescribed GA unchanged.

4. Vesper: How the four Losing Openings are punished

The full minimax values for X opening in columns 0,1,2,3,4 are -10,-10,1,-10,-10. Column 2 is the only winning opening. For each of the other four openings, O wins immediately by replying in column 2.

Using the assignment's bottom-based coordinates, O already occupies (0,0), (1,1) and (3,3). After X opens in column 0, 1, 3 or 4, O can drop into (2,2) and complete the diagonal. Nine cells remain empty after those two moves, giving utility -(1+9)=-10. X must fill that square first. With perfect play, opening in column 2 gives utility 1, meaning an X win with no empty cells left.

5. Jun: How much Search Effort is saved

Method	Value	Move	Nodes	Leaves
Full minimax	1	2	175, 789	64, 066
Depth-limited minimax, depth 4	-2	2	421	-
AB1, full depth	1	2	9064	-
AB2, full depth, middle first	1	2	8965	-
AB3, depth 4	-2	2	198	-

AB1 saves about 94.84% of full minimax nodes, and AB2 saves about 94.90%. Trying the middle first saves only another 99 nodes, or 1.09% of AB1. At depth four, AB3 saves about 52.97% compared with depth-limited minimax. Every count includes the root and terminal calls. The depth-four score of -2 is an estimate at the search horizon, so it does not contradict the full-search win.

Middle-first finds the necessary blocking move early, but good ordering must help throughout the tree to save much more work. This board is nearly full and asymmetric, with many restricted or terminal continuations. A fixed preference for the middle is therefore only a rough guide, which fits the small saving measured here.

Master Hyo's Tree: Left to Right Trace

Both visit orders give root value 9 and best move B. Left-to-right evaluates 39 leaves. Right-to-left evaluates 23, a saving of 16 leaves, or about 41.03%. Here is the manual left-to-right trace. When a node is pruned, its returned bound does not have to equal its exact minimax value.

At A1, A1a reads L1,L2,L3 and returns -27. A1b reads L4=-28 and cuts L5,L6 because -28 <= alpha=-27. A1c reads L7,L8,L9 and returns -20. A1d reads L10,L11,L12 and returns -15. A1 returns -15, so A sets beta=-15.

A2 receives (-inf,-15). Its children read L13,L14,L15 and return -29, then L16,L17 and return -22, then L18,L19 and return 0. A2 returns 0 at its beta cutoff, leaving A at -15. A3 receives the same window. A3a reads L20,L21,L22 and returns -15. Equality is enough to cut A3b,A3c,A3d. A returns -15 and R sets alpha=-15.

At B1, B1a reads L30,L31,L32 and returns -10. B1b reads L33,L34,L35 and returns -3. B1c reads L36,L37,L38 and returns -18. B1d reads L39,L40 and returns 26. B1 returns 26, so B sets beta=26.

B2 therefore receives exactly (alpha,beta)=(-15,26). B2a reads L41,L42 and returns 9, raising alpha to 9. B2b reads L43,L44=-3 and cuts L45 because -3 <= 9. B2c reads L46=-21 and cuts L47,L48 because -21 <= 9. B2 and B return 9. R raises alpha to 9 and selects B.

C1 receives (9,inf). C1a reads L49=5 and cuts L50,L51. C1b reads L52=-27 and cuts L53. C1c reads L54=-7 and cuts L55. C1 returns the bound 5. Since 5 <= alpha=9, C cuts all of C2. R returns 9 with move B.

So the 39 evaluated leaves in order are: L1, L2, L3, L4, L7, L8, L9, L10, L11, L12, L13, L14, L15, L16, L17, L18, L19, L20, L21, L22, L30, L31, L32, L33, L34, L35, L36, L37, L38, L39, L40, L41, L42, L43, L44, L46, L49, L52, L54.

Why Right to Left reaches fewer Leaves

C is searched first and returns -4. Within B, B2 comes before B1. B2c reads L48=-24 and cuts L47,L46 against alpha=-4. B2b reads L45,L44,L43 and returns -3. B2a reads L42,L41 and returns 9. B1 then starts with beta=9. Its first child, B1d, reads L40,L39 and returns 26, cutting the rest of B1. B returns 9 and raises the root alpha to 9.

Within A, A3 now comes first. It reads only L29, L27, L24,L23 and L22, returning the bound 7. Since 7 <= 9, A cuts A2 and A1 entirely. Cutting most of B1 and then most of A explains the large saving. An unpruned check gives A=-15, B=9 and C=-4, confirming the root value 9.